



2026 Google Awards for Machine Learning Research and Education with TPUs

Overview

Google is requesting proposals from faculty members at academic degree-granting institutions for the 2026 academic year. Google's goal for these awards is to support the advancement of machine learning through both foundational and applied research, as well as the design and execution of courses that feature Google TPUs. Proposals are welcome for research across all domains - from core ML systems to applied sciences - and for courses at any level, including undergraduate or postgraduate.

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Award Information

Selected proposals will receive a one-time unrestricted gift, compute resources (access to the latest generation of TPUs), technical mentorship, and Google Cloud credits:

- **Unrestricted gift:** A one-time unrestricted gift in the range of \$25,000 to \$100,000 USD, primarily intended to support or partially support graduate students (e.g., for

research or course development) or faculty. We are open to proposals outside this range depending on your project's scope and potential impact.

- **Access to TPUs:** In addition to the cash award, accepted proposals receive free-of-charge access to the latest generation of TPUs via Google Cloud, with allocations customized to your specific research or teaching needs.
- **Technical mentorship:** Faculty will have access to relevant product and engineering teams for guidance on best practices and technical troubleshooting.

Track 1: Research Awards

We are requesting proposals that leverage the capabilities of TPUs to accelerate important research across foundational ML, applied science, and open-source contributions.

Theme 1: Foundational ML

We aim to support research that pushes the boundaries of machine learning performance and scalability on TPUs:

- **ML Systems:** Designing Pallas kernels, compilers, distributed systems, or other techniques to maximize training and inference efficiency on TPUs.
 - We would love to see novel research [utilizing Pallas](#) to develop high-performance custom kernels (e.g., Sparse Attention, Quantization primitives, etc), and to develop open-source libraries.
 - Distributed Systems: Novel approaches to checkpointing, fault tolerance, etc for massive-scale training using MaxText or [PyTorch on TPUs](#).
- **Model innovation:** Advancing techniques for pre-training, post-training ([Tunix](#)), and inference optimization ([vLLM on TPUs](#)), as well as research into reinforcement learning and architectures

Theme 2: Applied Science

We are eager to support applied fields where high-performance computing via TPUs can unlock new discoveries:

- **Healthcare & Life Sciences:** Projects in generative biology, biomolecular modeling, and multimodal clinical foundation models or applications.
- **Climate and crisis resilience:** Projects that leverage machine learning to improve global sustainability and disaster response, including food security, agricultural optimization, and wildfire mitigation.
- **Accessibility:** Projects to develop inclusive technologies that remove barriers for individuals with disabilities through innovations in computer vision, speech recognition, assistive AI, or related efforts.

- **Computer-Aided Engineering (CAE) & Science:** Research utilizing physics-informed neural networks, neural operators, digital twins, and large-scale weather or climate modeling workflows using [JAX](#)/TPUs.
- **Note:** We welcome proposals from all applied domains where TPU compute can make a difference.

Theme 3: Open Source Contributions

We are looking to support contributors whose primary output is code and systems artifacts. We encourage proposals from researchers and engineers building the foundational tools that the broader academic community relies on for TPUs:

- **Ecosystem Contributions:** Proposals that contribute directly to the open-source community, such as developing new algorithms for Tux, enhancing the MaxText ecosystem, or building tools that benefit the broader TPU developer community.
- **Systems Researchers:** Implementing novel distributed training algorithms (e.g., new sharding strategies or fault-tolerance mechanisms) on frameworks like MaxText or PyTorch.
- **Kernel Engineers:** Developing custom operations, such as hardware-aware implementations of Ring Attention, Mamba/SSMs, or FP8 primitives.
- **Model Architects:** Porting and optimizing frontier architectures (e.g., Mixture-of-Depths, Diffusion Transformers) to run efficiently on XLA-based stacks.
- **Tool Builders:** Creating profilers, debuggers, or visualization tools that clarify distributed system performance and lower the barrier to entry for other researchers.

Track 2: Education Awards

We aim to empower faculty to prepare the next generation of AI practitioners by integrating TPUs into the curriculum. We are requesting proposals for educational initiatives that bring TPUs into the curriculum, ranging from full-semester course designs to modular integrations - such as hands-on workshops, specialized labs, or lecture series within existing courses. Our goal is to equip students with the practical expertise required to build, use, and deploy modern ML systems on TPUs, in either basic or applied courses.

Focus Area 1: LLM & Systems Courses

We support the development of curriculum that delves into the engineering and systems challenges of modern machine learning:

- **Large Language Models (LLMs):** Content focused on the architecture, pre-training, fine-tuning, and efficient serving of LLMs (focusing on model internals rather than API usage).

- **Distributed Systems & Scalability:** Teaching the fundamentals of distributed training, including data parallelism, model parallelism, and sharding techniques required to run large-scale workloads on TPUs.
- **Hardware-Aware ML:** Courses that explore lower-level optimization, compiler stacks (e.g., XLA), or custom kernel development (e.g., Pallas) to maximize hardware utilization.

Focus Area 2: Applied Science Courses

Machine learning is now an essential tool for scientific discovery and applied science. We welcome proposals for courses outside of traditional Computer Science departments:

- **"ML for Science" Domains:** Courses tailored for applied fields (e.g., *AI for Healthcare*, *Deep Learning for Physics*, *Computational Biology*, *Climate Modeling*) that teach students how to apply high-performance computing to domain-specific datasets.
- **Capstone Projects:** Project-based courses where student teams build and deploy end-to-end ML applications, bridging the gap between theory and industrial practice.

Focus Area 3: Workshops, Labs & Modular Curriculum

Recognizing that full-semester syllabus updates may not be feasible for the upcoming year, we are open to proposals for flexible, shorter-term educational formats:

- **Plug-and-Play Labs:** The development of single, high-impact lab assignments (e.g., "Fine-tuning Gemma with Tunix on TPUs") that can be inserted into an existing ML course without restructuring the entire syllabus.
- **Lecture series or guest lectures:** Lectures or guest lectures on TPUs that can be incorporated into an existing ML course.
- **Workshops:** Short-form, intensive sessions designed to get students up and running with TPUs quickly.

Recommended software and technologies

To help you hit the ground running, we recommend leveraging the following software ecosystems, which are highly optimized for TPUs.

- **[MaxText](#):** A high-performance, large-scale LLM training and inference framework.
- **[Tunix](#):** A post-training and reinforcement learning library designed for scalability and performance on TPUs.

- [The Scaling ML Book](#): This book aims to demystify the science of scaling language models: how TPUs (and GPUs) work and how they communicate with each other, and how LLMs run on real hardware.
- [vLLM on TPUs](#): For high-throughput / low-latency inference serving, we recommend using vLLM, which now has dedicated TPU support.
- [JAX](#): A Python library for accelerator-oriented array computation and program transformation, widely used for high-performance ML research. See also: [JAX-LLM-Examples](#) and [Bonsai](#), two collections of models implemented in JAX.
- [PyTorch on TPUs](#): We are actively developing the PyTorch TPU stack to ensure a seamless experience. For background on our roadmap and current status, see our PyTorch on TPU RFC.
- [Keras 3](#): Easy to use, multi-framework machine learning that works with a JAX, PyTorch, or TensorFlow backend.
- [Gemma](#): A family of lightweight, state-of-the-art open models.

Other software is welcome too, and we are happy to discuss options with you to find the best fit for your project (email us at tpu-rfp@google.com)

FAQ

Who can apply? These awards are open to faculty members at degree-granting colleges and universities in any country, except those with sanctions imposed by the United States Office of Foreign Assets Control (OFAC), as well as NGOs / IGOs (intergovernmental or international) and non-profit organizations.

Can PhD students apply? Yes! PhD students are welcome to apply. Please include a faculty supervisor on your application.

Are these awards given to individuals or the university? Awards are given as an unrestricted gift to the university. Please note that Google does not pay overhead on gifts.

How is this program different from the TPU Research Cloud? [The TPU Research Cloud](#) is a separate program that supports researchers by providing free access to TPUs. While both programs provide free access to TPUs, these awards are intended to support research or teaching, and include a one-time unrestricted gift. We work closely with the TRC team and may recommend applying to both programs depending on the scale of your compute needs.

Will I receive feedback on my proposal if it is not selected? Due to the high volume of applications we receive, we are unable to provide individual feedback on proposals.

Review Criteria

Proposals will be evaluated based on their relevance to TPU ecosystem, technical merit and innovation, feasibility, and potential for open-source impact. For education tracks, we consider the quality and potential reach of the proposed course materials.

Submit your application

Please write your proposal as a 1-2 page PDF and submit it through [this form](#).

To ensure a clear path to success, proposals must include defined milestones:

- **For Research:** Specify the research topic, your related work, intended publication venues (e.g., NeurIPS, ICML), and expected submission dates, and/or expected open-source contributions.
- **For Teaching:** Specify the course format, expected enrollment, recording plans, and dates for when the class/workshop will be published.

Please include a funding request and a budget outlining how the funds will be used. These awards are competitive. The program committee will determine the final funding amount.

If you know the type and number of TPUs that would be helpful for your project, please include that in your proposal. If you don't, that's okay, and we will work with you on an ideal allocation (you can also start small and ramp up as needed).

If your proposal is accepted, we ask that faculty send a brief quarterly update on their progress. This visibility allows us to provide proactive support, answer technical questions, share opportunities, and connect you with relevant engineering resources and future opportunities.

Finally, we ask that you open-source your project code under an Apache 2.0 or MIT license to share it with the community.

Application deadline. Decisions are rolling, and will accept applications until June 30th, 2026. You will hear back in 4-6 weeks after submitting your application.

Questions?

If you have any questions about the RFP or would like to discuss your proposal, please reach out to us at tpu-rfp@google.com.